

Linux Class 7 - Practice Task

Practising hard links, soft links and the more command

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This is the practice I did after class 7. I made a folder called practice7.

What is a link

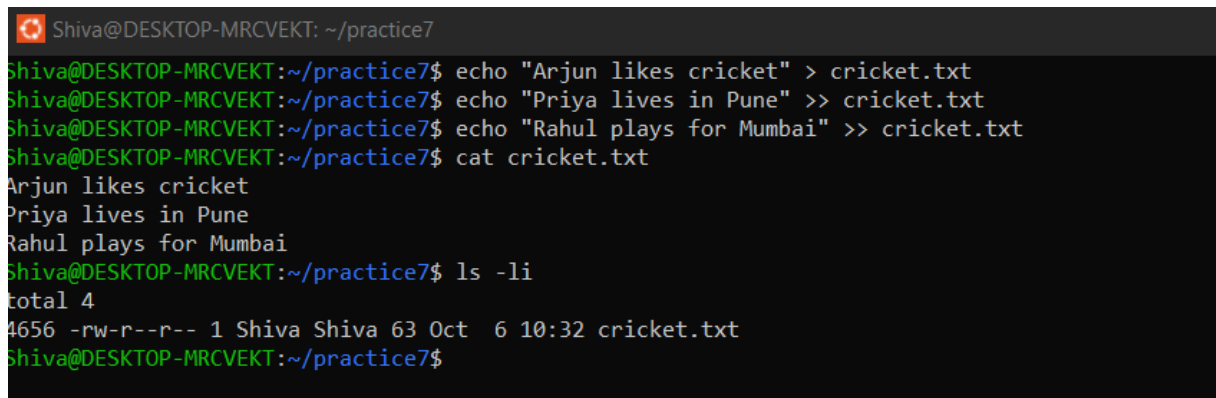
From the notes: a link is a pointer to a file or a folder. It gives me one more way to reach the same data, without making a second copy of it. There are two kinds, hard link and soft link. I made a small file and tried both.

A word I had to learn first is inode. An inode is a small record that Linux keeps for every file. It stores things like the size and the owner. Each one has a number, and ls -li shows that number in the first column.

Making the file

My file is cricket.txt. I put 3 lines in it with echo. The first echo has one > and the next two have >>, because one arrow makes a new file and two arrows add to it.

```
echo "Arjun likes cricket" > cricket.txt
echo "Priya lives in Pune" >> cricket.txt
echo "Rahul plays for Mumbai" >> cricket.txt
cat cricket.txt
ls -li
```

A screenshot of a terminal window with a dark background. The prompt is 'Shiva@DESKTOP-MRCVEKT: ~/practice7'. The user enters several commands: 'echo "Arjun likes cricket" > cricket.txt', 'echo "Priya lives in Pune" >> cricket.txt', 'echo "Rahul plays for Mumbai" >> cricket.txt', 'cat cricket.txt', and 'ls -li'. The output of 'cat' shows three lines of text. The output of 'ls -li' shows a single line: '4656 -rw-r--r-- 1 Shiva Shiva 63 Oct 6 10:32 cricket.txt'.

```
Shiva@DESKTOP-MRCVEKT: ~/practice7
Shiva@DESKTOP-MRCVEKT:~/practice7$ echo "Arjun likes cricket" > cricket.txt
Shiva@DESKTOP-MRCVEKT:~/practice7$ echo "Priya lives in Pune" >> cricket.txt
Shiva@DESKTOP-MRCVEKT:~/practice7$ echo "Rahul plays for Mumbai" >> cricket.txt
Shiva@DESKTOP-MRCVEKT:~/practice7$ cat cricket.txt
Arjun likes cricket
Priya lives in Pune
Rahul plays for Mumbai
Shiva@DESKTOP-MRCVEKT:~/practice7$ ls -li
total 4
4656 -rw-r--r-- 1 Shiva Shiva 63 Oct  6 10:32 cricket.txt
Shiva@DESKTOP-MRCVEKT:~/practice7$
```

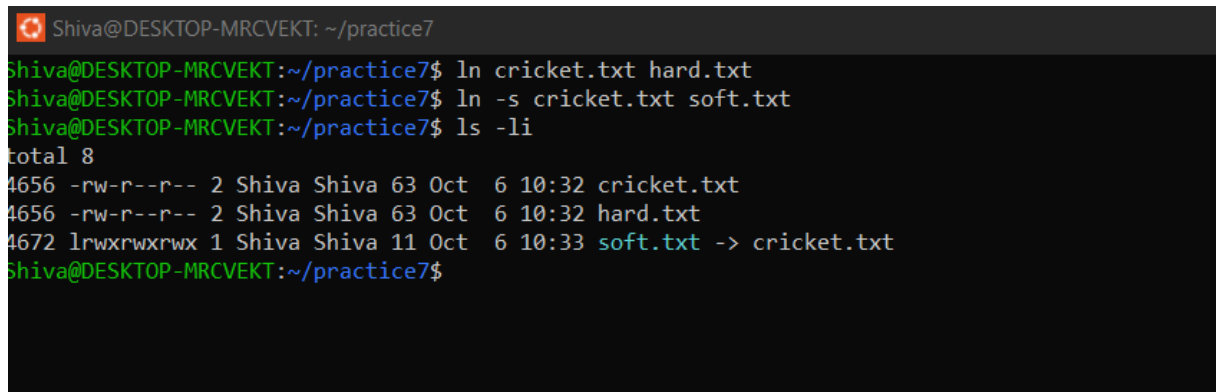
Screenshot 1: making cricket.txt and looking at its inode number.

The number at the start of the ls line, 4656, is the inode number. The 1 after the permissions is how many names the file has right now. It has only one name, so it says 1. I will watch this number change.

Making a hard link and a soft link

The hard link is ln and then the file name and then the new name. The soft link is the same but with -s. I made one of each.

```
ln cricket.txt hard.txt
ln -s cricket.txt soft.txt
ls -li
```



```
Shiva@DESKTOP-MRCVEKT: ~/practice7
Shiva@DESKTOP-MRCVEKT:~/practice7$ ln cricket.txt hard.txt
Shiva@DESKTOP-MRCVEKT:~/practice7$ ln -s cricket.txt soft.txt
Shiva@DESKTOP-MRCVEKT:~/practice7$ ls -li
total 8
4656 -rw-r--r-- 2 Shiva Shiva 63 Oct  6 10:32 cricket.txt
4656 -rw-r--r-- 2 Shiva Shiva 63 Oct  6 10:32 hard.txt
4672 lrwxrwxrwx 1 Shiva Shiva 11 Oct  6 10:33 soft.txt -> cricket.txt
Shiva@DESKTOP-MRCVEKT:~/practice7$
```

Screenshot 2: a hard link and a soft link made from cricket.txt.

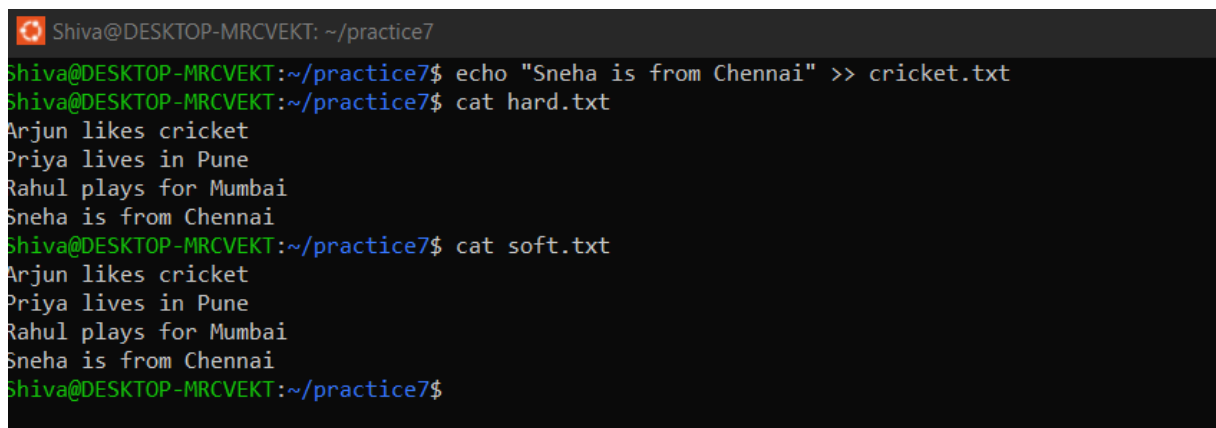
Now I could see the difference in the ls output.

- cricket.txt and hard.txt both have inode 4656. So they are the same file with two names. The number after the permissions changed from 1 to 2 for both of them.
- soft.txt has a different inode, 4672. It also starts with an l instead of a dash, and there is an arrow showing soft.txt -> cricket.txt. Its size is 11, which is just the 11 letters of the name cricket.txt. So a soft link only stores the name of the file it points to.

Changing the original

The first note says if I change the original, I should see it in the link. I added one more line to cricket.txt and then read both links.

```
echo "Sneha is from Chennai" >> cricket.txt
cat hard.txt
cat soft.txt
```



```
Shiva@DESKTOP-MRCVEKT: ~/practice7
Shiva@DESKTOP-MRCVEKT:~/practice7$ echo "Sneha is from Chennai" >> cricket.txt
Shiva@DESKTOP-MRCVEKT:~/practice7$ cat hard.txt
Arjun likes cricket
Priya lives in Pune
Rahul plays for Mumbai
Sneha is from Chennai
Shiva@DESKTOP-MRCVEKT:~/practice7$ cat soft.txt
Arjun likes cricket
Priya lives in Pune
Rahul plays for Mumbai
Sneha is from Chennai
Shiva@DESKTOP-MRCVEKT:~/practice7$
```

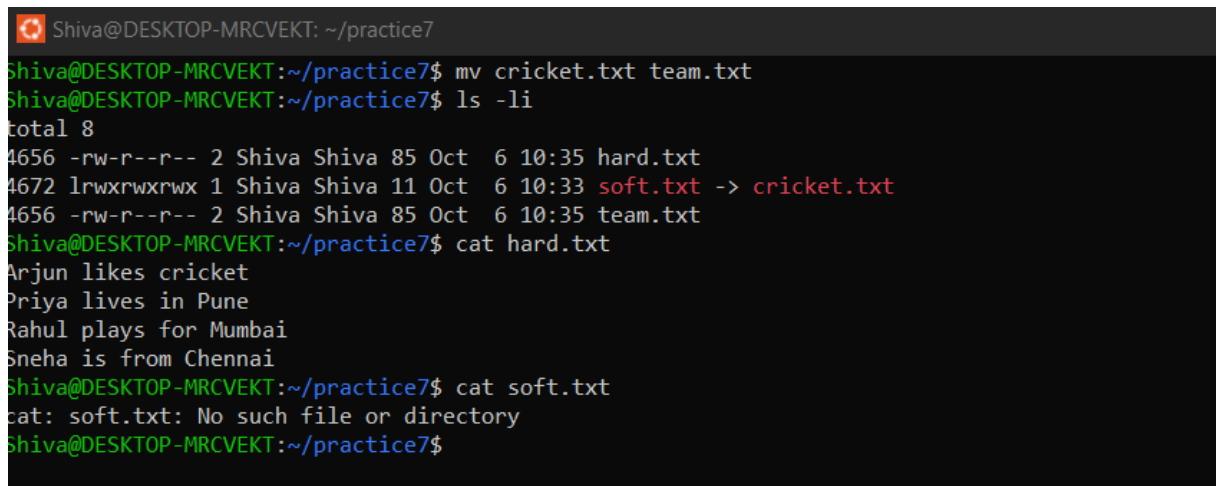
Screenshot 3: the new line shows up when I read both links.

Both of them showed Sneha is from Chennai. So for this test the two kinds of link behave the same.

Renaming the original

Now the part where they are different. I renamed cricket.txt to team.txt with mv, then looked at the folder and tried to read both links.

```
mv cricket.txt team.txt
ls -li
cat hard.txt
cat soft.txt
```



```
Shiva@DESKTOP-MRCVEKT: ~/practice7
Shiva@DESKTOP-MRCVEKT:~/practice7$ mv cricket.txt team.txt
Shiva@DESKTOP-MRCVEKT:~/practice7$ ls -li
total 8
4656 -rw-r--r-- 2 Shiva Shiva 85 Oct  6 10:35 hard.txt
4672 lrwxrwxrwx 1 Shiva Shiva 11 Oct  6 10:33 soft.txt -> cricket.txt
4656 -rw-r--r-- 2 Shiva Shiva 85 Oct  6 10:35 team.txt
Shiva@DESKTOP-MRCVEKT:~/practice7$ cat hard.txt
Arjun likes cricket
Priya lives in Pune
Rahul plays for Mumbai
Sneha is from Chennai
Shiva@DESKTOP-MRCVEKT:~/practice7$ cat soft.txt
cat: soft.txt: No such file or directory
Shiva@DESKTOP-MRCVEKT:~/practice7$
```

Screenshot 4: after the rename the hard link still works and the soft link does not.

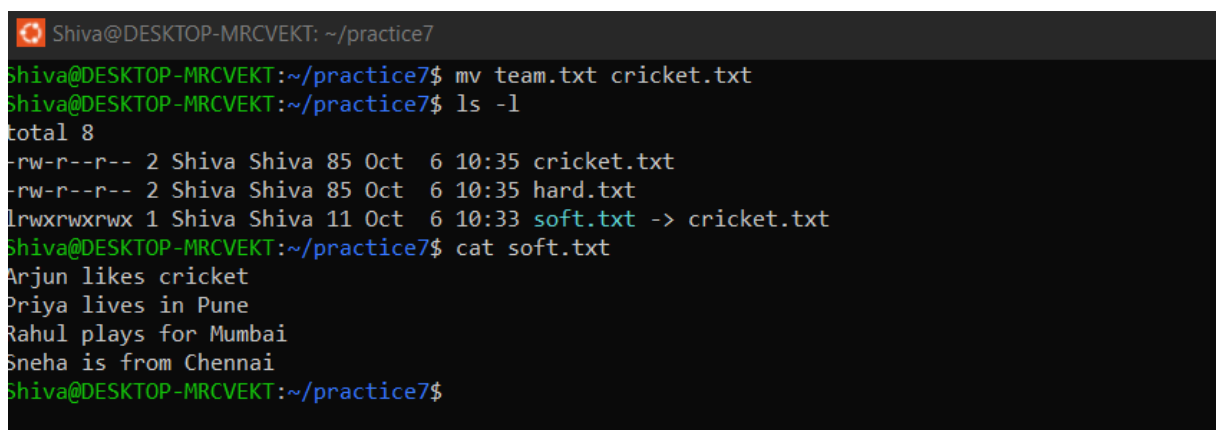
hard.txt still worked and showed all 4 lines. In ls, hard.txt and team.txt still have the same inode, 4656, so they are the same file. A rename only changes one of the names.

soft.txt did not work. cat said No such file or directory, and ls showed the name in red. That is because the soft link only knows the old name cricket.txt, and that name is not there any more. The link is still sitting there but it points to nothing.

Putting the name back

I wanted to check my idea that the soft link only remembers the name. So I renamed team.txt back to cricket.txt.

```
mv team.txt cricket.txt
ls -li
cat soft.txt
```



```
Shiva@DESKTOP-MRCVEKT: ~/practice7
Shiva@DESKTOP-MRCVEKT:~/practice7$ mv team.txt cricket.txt
Shiva@DESKTOP-MRCVEKT:~/practice7$ ls -li
total 8
-rw-r--r-- 2 Shiva Shiva 85 Oct  6 10:35 cricket.txt
-rw-r--r-- 2 Shiva Shiva 85 Oct  6 10:35 hard.txt
lrwxrwxrwx 1 Shiva Shiva 11 Oct  6 10:33 soft.txt -> cricket.txt
Shiva@DESKTOP-MRCVEKT:~/practice7$ cat soft.txt
Arjun likes cricket
Priya lives in Pune
Rahul plays for Mumbai
Sneha is from Chennai
Shiva@DESKTOP-MRCVEKT:~/practice7$
```

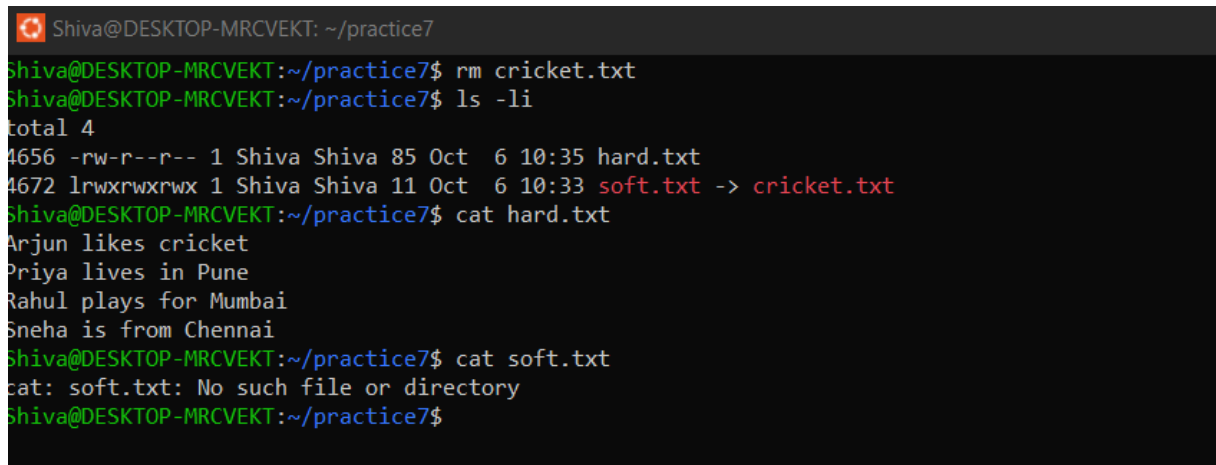
Screenshot 5: with the old name back, the soft link works again.

The soft link worked again, and the red colour went away. So my idea was right. The soft link is just the name of a file, so when that name is back it finds the file again.

Deleting the original

The last test from the notes is to delete the original file.

```
rm cricket.txt
ls -li
cat hard.txt
cat soft.txt
```

A screenshot of a terminal window with a dark background. The prompt is 'Shiva@DESKTOP-MRCVEKT: ~/practice7'. The user enters 'rm cricket.txt'. Then they enter 'ls -li', which shows two files: '4656 -rw-r--r-- 1 Shiva Shiva 85 Oct 6 10:35 hard.txt' and '4672 lrwxrwxrwx 1 Shiva Shiva 11 Oct 6 10:33 soft.txt -> cricket.txt'. The 'soft.txt' entry is highlighted in red. Then the user enters 'cat hard.txt', which displays four lines of text: 'Arjun likes cricket', 'Priya lives in Pune', 'Rahul plays for Mumbai', and 'Sneha is from Chennai'. Finally, the user enters 'cat soft.txt', which results in an error message: 'cat: soft.txt: No such file or directory'.

Screenshot 6: the original is gone, the hard link still has the data.

hard.txt still showed all 4 lines. Its link count went back down to 1. The data is not gone as long as one name is left. I think of rm as only taking away one name, and the file really goes only when the last name is removed.

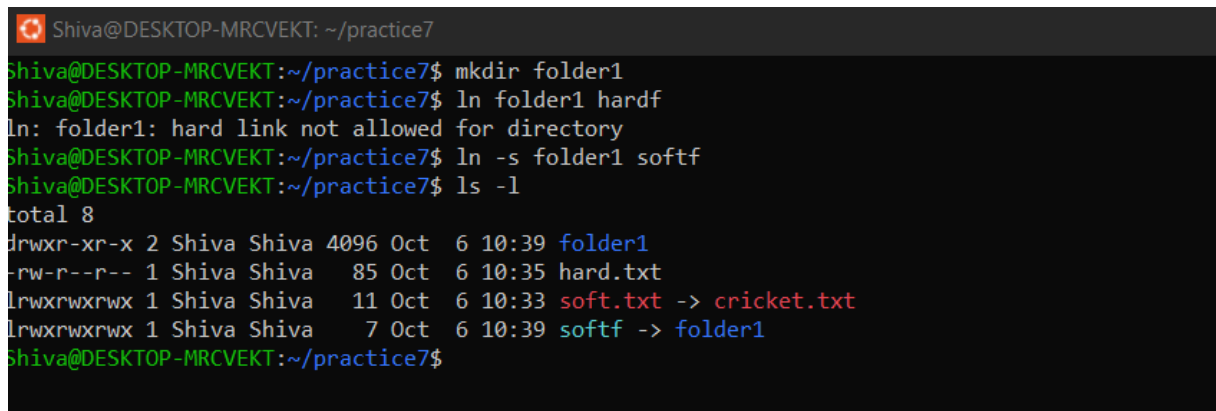
soft.txt broke again, with the same error as before and the red name in ls.

Thing I tried	Hard link (hard.txt)	Soft link (soft.txt)
Inode number	Same as the original (4656)	Its own number (4672)
Looks like in ls -l	A normal file, no arrow	Starts with l and has -> in it
I added a line to the original	Showed up in the link	Showed up in the link
I renamed the original	Still worked	Broke, cat said no such file
I put the old name back	Nothing to fix, it never broke	Worked again
I deleted the original	Still worked, all 4 lines there	Broke, red name in ls
Link on a folder	Not allowed	Worked

Links for folders

Task 4 in class asked if we can make links for folders. That is not in the notes but it is easy to try, so I did it to find out.

```
mkdir folder1
ln folder1 hardf
ln -s folder1 softf
ls -l
```

A terminal window titled 'Shiva@DESKTOP-MRCVEKT: ~/practice7'. The user enters the following commands: 'mkdir folder1', 'ln folder1 hardf', 'ln -s folder1 softf', and 'ls -l'. The output of 'ls -l' shows a directory listing with permissions, owner, size, date, and file names. The 'softf' file is shown as a symbolic link pointing to 'folder1'.

```
Shiva@DESKTOP-MRCVEKT: ~/practice7
Shiva@DESKTOP-MRCVEKT:~/practice7$ mkdir folder1
Shiva@DESKTOP-MRCVEKT:~/practice7$ ln folder1 hardf
ln: folder1: hard link not allowed for directory
Shiva@DESKTOP-MRCVEKT:~/practice7$ ln -s folder1 softf
Shiva@DESKTOP-MRCVEKT:~/practice7$ ls -l
total 8
drwxr-xr-x 2 Shiva Shiva 4096 Oct  6 10:39 folder1
-rw-r--r-- 1 Shiva Shiva   85 Oct  6 10:35 hard.txt
lrwxrwxrwx 1 Shiva Shiva   11 Oct  6 10:33 soft.txt -> cricket.txt
lrwxrwxrwx 1 Shiva Shiva    7 Oct  6 10:39 softf -> folder1
Shiva@DESKTOP-MRCVEKT:~/practice7$
```

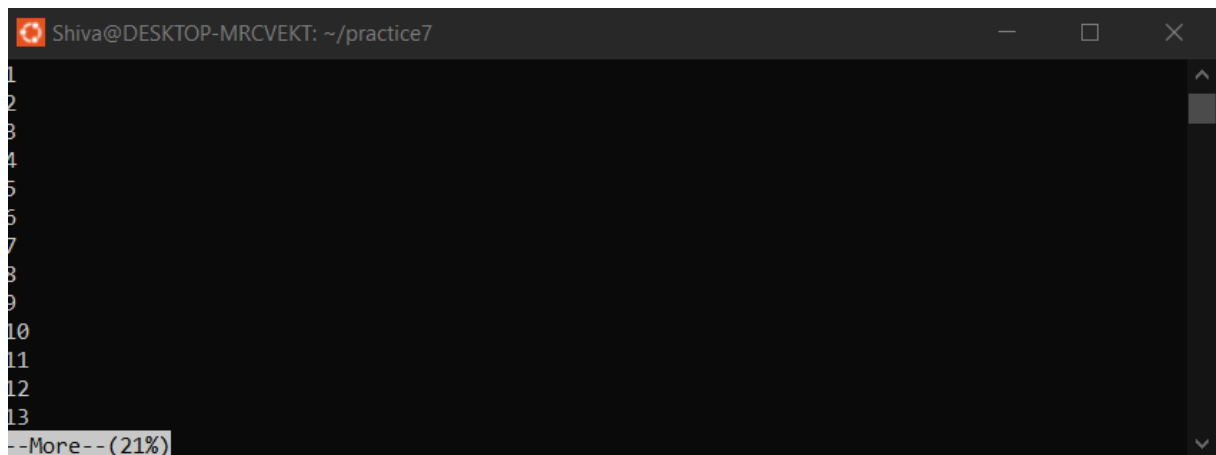
Screenshot 7: the hard link on a folder is refused, the soft link works.

The hard link gave an error, ln: folder1: hard link not allowed for directory, and no hardf was made. The soft link softf worked and shows softf -> folder1 in blue like a folder. So for folders only a soft link can be used.

The more command

The notes say more shows a file one page at a time. A short file will not show this, so I made a file with 50 lines using seq, which just counts. I made my terminal window shorter so the pages are small.

```
seq 1 50 > numbers.txt
more numbers.txt
```

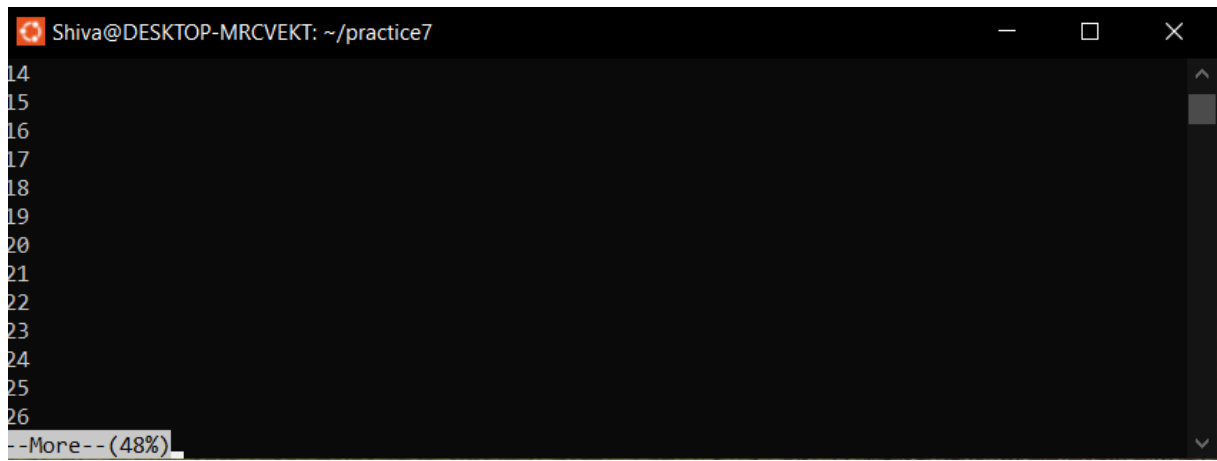
A terminal window titled 'Shiva@DESKTOP-MRCVEKT: ~/practice7'. The user has run 'seq 1 50 > numbers.txt' and 'more numbers.txt'. The terminal shows the first 13 lines of the file, numbered 1 to 13. At the bottom, it says '--More-- (21%)'.

```
Shiva@DESKTOP-MRCVEKT: ~/practice7
1
2
3
4
5
6
7
8
9
10
11
12
13
--More-- (21%)
```

Screenshot 8: more shows only the first page and waits.

more cleared the screen and showed the first lines, 1 to 13. At the bottom it says --More--(21%). That means I have seen 21 percent of the file. It waits there until I press a key.

The notes say to press the down arrow to see the next content. I pressed the space bar instead and it showed the next page, which is the next screenshot.

A terminal window titled "Shiva@DESKTOP-MRCVEKT: ~/practice7" with standard window controls. The terminal displays a vertical list of numbers from 14 to 26 on the left side. At the bottom left, there is a status bar that reads "--More-- (48%)". A vertical scrollbar is visible on the right side of the terminal window, indicating the current scroll position.

Screenshot 9: after pressing space, the next page, now 48 percent.

Now it shows 14 to 26 and the percentage is 48. The notes say q is for quit, so that is how to leave more before the end.